Nature of Ultrasound Waves

Definition

Ultrasound waves are mechanical longitudinal sound waves with frequencies higher than 20 kHz, which is above the range of human hearing. Therefore, they are inaudible to humans.
They require a material medium (such as air, water, or solids) to propagate and cannot travel through a vacuum.

Basic Properties

1. Frequency

- Higher than 20 kHz
- Can reach several MHz in applications
- Higher frequency leads to higher precision and detail

2. Wavelength

- Very short compared to ordinary sound waves
- Short wavelength allows detection of fine details

3. Propagation Speed

- Depends on the medium
- Slowest in gases, faster in liquids, and fastest in solids

4. Mechanical Nature

- Depends on vibrations of particles in a medium
- Travels as **compressions and rarefactions**
- Cannot propagate in vacuum

5. Directionality

- Can be focused into narrow beams
- Improves with higher frequency

6. Absorption

- Loses part of its energy during propagation
- Energy is converted into heat
- Increases with frequency

7. Reflection and Refraction

- **Reflection**: occurs when the wave hits a boundary between two different media
- **Refraction**: occurs when the wave changes direction while passing into another medium

8. Scattering

- Happens due to small particles or irregularities in the medium
- Causes the wave to spread in different directions

9. Intensity

- Represents the energy carried by the wave
- Increases with the strength of the source

□ Piezoelectric Effect and Ultrasonic Transducers

1. Piezoelectric Effect

The piezoelectric effect is a physical phenomenon in certain crystalline materials where mechanical and electrical energy are converted into each other:

Connection to Ultrasound

The piezoelectric effect is the fundamental principle behind generating and detecting ultrasound waves.

2. Ultrasonic Transducers

Definition

An ultrasonic transducer is a device that converts electrical energy into ultrasound waves and vice versa using the piezoelectric effect.

Working Principle

1. Transmission Mode

- An alternating electrical signal is applied to a piezoelectric crystal
- The crystal rapidly vibrates
- These vibrations produce ultrasound waves in the surrounding medium

2. Reception Mode

- Incoming ultrasound waves strike the crystal
- The crystal vibrates due to the wave energy
- These vibrations are converted into an electrical signal

Main Components

- Piezoelectric crystal (such as quartz or PZT ceramics)
- Protective outer casing
- Electrical electrodes
- Matching layers for efficient wave transmission

Applications

1. Medicine

- Ultrasound imaging
- Fetal monitoring
- Internal organ examination

2. Industry

- Non-destructive testing of materials
- Thickness measurement

3. Navigation

- Sonar systems in ships and submarines

Beam Formation and Focusing

Introduction

In ultrasound systems, sound is not transmitted randomly. Instead, it is precisely controlled using two key techniques:

- **Beam Formation**
- **Beam Focusing**

Both techniques rely on controlling **wave interference** using arrays of piezoelectric elements.

1. *Beam Formation*

Basic Idea

Beam formation is the process of **directing sound waves in a specific direction** instead of allowing them to spread in all directions.

Physical Principle

An **array of transmitting elements** is used.
Each element emits a wave, but not at exactly the same time.

Control parameters include:

- **Phase**
- **Time delay**
- **Amplitude**

What happens physically?

Due to interference:

- In the desired direction → Constructive interference → strong beam
- In other directions → Destructive interference → weak signal

Applications

- Steering ultrasound beams inside the body
- Scanning different directions in imaging
- Radar and sonar systems

2. Beam Focusing

Basic Idea

Beam focusing is the process of concentrating waves at a single point in space, called the focal point.

Physical Principle

Each element in the array is controlled so that:

- Waves travel different distances
- But arrive at the same point at the same time

This is called: Time Delay Focusing

What happens physically?

At the focal point:

- All waves align perfectly
- Energy is added together
- Intensity becomes very high

Away from the focal point:

- Waves are not fully aligned
- Energy is weaker

Applications

- High-resolution medical imaging
- Precise targeting of tissues
- Lithotripsy (kidney stone treatment)

Ultrasound Propagation in Media

Ultrasound in biological media

Ultrasound waves are high-frequency sound waves (above the human hearing range). They are widely used in the body because they can travel through tissues with relatively low damage when used at appropriate energy levels.

How do they propagate in biological tissues?

- They travel well through fluids and soft tissues (such as muscle and blood)
- They are partially reflected at boundaries between different tissues (e.g., muscle/bone)
- They are partially absorbed and converted into a small amount of heat

Why do tissues respond differently?

Each tissue has different physical properties such as:

- Density
- Speed of sound within it
- Acoustic impedance

Because of this:

- Some waves pass through easily
- Some are strongly reflected
- Some are more strongly absorbed

What happens inside the body?

When ultrasound is used:

- Short pulses of sound waves are sent into the body
- Echoes are reflected from different tissues

- These echoes are analyzed to form images (as in medical ultrasound imaging)

Applications:

- Medical imaging (ultrasound imaging)
- Monitoring a fetus during pregnancy
- Breaking kidney stones (lithotripsy)
- Focused ultrasound therapy

Reflection in Biological Media

Reflection is the process in which part of an ultrasound wave returns back when it hits a boundary between two different tissues inside the body.

Why does reflection occur?

It happens because different tissues have different **acoustic impedance** (a physical property related to density and sound speed). When there is a mismatch in acoustic impedance:

- Part of the wave is transmitted forward
- Part of the wave is reflected back

Where does it occur?

Reflection happens at interfaces between different tissues, such as:

- muscle ↔ fat
- muscle ↔ bone
- fluid ↔ soft tissue

What is produced?

- Reflected waves form an **echo**
- These echoes are detected by the ultrasound machine
- They are used to create medical images

Types of reflection:

- Strong reflection: occurs when the difference between tissues is large (e.g., bone and muscle)
- Weak reflection: occurs when tissues are similar